

Case study

COURSE CODE: CSA0702

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1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	deeply In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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CASE STUDY 1 – Video Conferencing Issue

1. Analysis of the Problem (Transport vs Application Layer)

The issue is mainly caused by the **transport protocol (UDP)** and partially by the **application layer buffering**.

- Video conferencing applications usually use **UDP** because it provides faster data transmission without waiting for acknowledgments.
- However, **UDP does not guarantee reliability**, leading to **packet loss (8%) and jitter (180 ms)**.
- The application layer may also contribute due to **improper buffer management**, which causes delay and frozen frames.

Therefore, the problem is a **combined effect**, but mainly due to **UDP limitations in handling packet loss and jitter**.

2. Justification Based on Real-Time Multimedia Requirements

Real-time applications like video conferencing require:

- **Low latency** → Data must reach quickly
- **Low jitter** → Smooth playback
- **Tolerance to some loss** → Small losses acceptable

Comparison:

- **TCP**: Reliable but slow (retransmissions cause delay)
- **UDP**: Fast but unreliable

Since the system shows **distortion and delay**, it indicates that:

- **UDP is used (for speed)**, but
- **Network conditions (loss + jitter)** are degrading quality

Thus, UDP is suitable, but **needs optimization**.

3. Recommendations for Improvement

Protocol-Level Improvements:

- Use RTP (Real-Time Transport Protocol) over UDP
- Implement Forward Error Correction (FEC) to recover lost packets
- Apply QoS (Quality of Service) to prioritize video traffic

Application-Level Improvements:

- Use adaptive bitrate streaming
- Improve jitter buffering
- Use packet loss concealment techniques

Network Improvements:

- Reduce congestion
- Increase bandwidth stability

CASE STUDY 2 – DNS Delay Problem

1. Reasons for High DNS Resolution Time (800 ms)

- Centralized DNS servers → Far from users
- No caching → Every request processed again
- High network latency
- Improper TTL values → Frequent lookups
- Server overload

2. Proposed DNS Architecture

To reduce lookup time below 50 ms:

- Anycast DNS
 - Multiple DNS servers globally
 - Requests routed to nearest server
- DNS Pre-fetching
 - Browser fetches DNS before user clicks
- TTL Optimization
 - Increase TTL for frequently accessed domains
- Distributed DNS Servers
 - Place servers in multiple regions

3. Advantages and Limitations

Advantages:

- Faster response (<50 ms)
- High availability
- Load distribution
- Reduced latency

Limitations:

- Complex setup
- Cost of infrastructure
- Cache inconsistency possible

CASE STUDY 3 – Centralized DNS Issue

4. Impact of Centralized DNS Architecture

- High latency (800 ms)
- Single point of failure
- Server overload
- Poor scalability
- Slow application response

5. Distributed DNS Solution Design

- Deploy **multiple DNS servers globally**
- Use **Anycast routing**
- Implement **DNS caching** at ISP and client level
- Add **load balancing**

Working:

- User request → nearest DNS server → faster response

6. Effect of TTL, Anycast, and Caching

- **TTL (Time to Live):**
 - High TTL → fewer queries, faster performance
 - Low TTL → more updates, less caching
- **Anycast Routing:**
 - Sends request to nearest server
 - Reduces latency

- **DNS Caching:**

- Stores previous results
- Reduces repeated lookups

Together, they improve performance and fault tolerance.

CASE STUDY 4 – Stock Trading Platform (TCP Issues)

1. TCP Factors Causing Latency Increase

- **Flow Control:**

- Limits data transmission speed
- Causes delay when buffer is full

- **Nagle's Algorithm:**

- Combines small packets → increases delay
- Not suitable for real-time systems

- **SYN Queue Limitations:**

- Too many connection requests
- Leads to delays in connection setup

2. Effect on High-Frequency Transactions

- Delay in order processing
- Slow trade confirmations
- Reduced system performance
- Customer dissatisfaction

3. Recommended TCP Improvements

- Disable Nagle's Algorithm (TCP_NODELAY)
 - Increase TCP buffer size
 - Optimize SYN queue length
 - Use TCP Fast Open
 - Reduce retransmissions with better congestion control
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